

Disproof of Conjecture 00000007672: $\gcd(F_{55}, F_{11}) = 1$ and $\gcd(F_6, F_3) = 1$ in $\mathbb{Z}[q]$

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Abstract

Conjecture 00000007672 asserts that for the q -Fibonacci polynomials $F_n(q) = \sum_j \begin{bmatrix} n-1-j \\ j \end{bmatrix}_q q^{j^2}$, the greatest common divisor $\gcd(F_n, F_m)$ in $\mathbb{Z}[q]$ equals $F_{\gcd(n,m)}$ times a correction factor. We give two counterexamples. First, $\gcd(F_6, F_3) = 1$ in $\mathbb{Z}[q]$ while $F_{\gcd(6,3)} = F_3 = 1 + q$. Second, and robust under every reading in which the exceptional set E depends only on primes $p \equiv \pm 2 \pmod{5}$: $\gcd(F_{55}, F_{11}) = 1$ while $F_{\gcd(11,55)} = F_{11}$ is a monic polynomial of degree 25 with no cyclotomic factor. Here all prime factors of 11 and 55 are 5 or 11, none $\equiv \pm 2 \pmod{5}$. Lean 4 (core library only) proves the $(6, 3)$ counterexample completely, including the gcd computation, and proves that F_{11} does not divide F_{55} . The facts that $\gcd(F_{55}, F_{11}) = 1$ and that F_{11} has no cyclotomic factor are certified computationally. Finally, $F_d \nmid F_{5d}$ for every odd $d \geq 11$ with $d \neq 15$, so the failures cannot be confined to finitely many indices either.

1 The conjecture

The statement (English version) reads:

Definition: The q -Fibonacci polynomials are $F_n(q) = \sum [n-1-j \text{ choose } j]_q q^{j^2}$, with Gaussian binomial coefficients. *Conjecture:* $\gcd(F_n(q), F_m(q))$ in $\mathbb{Z}[q]$ equals $F_{\gcd(n,m)}(q)$ times a correction factor from a finite exceptional set E depending only on primes $p \equiv \pm 2 \pmod{5}$, and the count of $E \cap [1, x]$ is $O(\log \log x)$.

The Chinese version says the same thing. The conjecture is a conjunction, so refuting its first clause refutes it. That clause is:

$$\gcd(F_n, F_m) = F_{\gcd(n,m)} \cdot c_{n,m} \quad \text{for some polynomial } c_{n,m}. \quad (1)$$

The “correction factor” multiplies a polynomial and is supposed to yield the gcd, which lies in $\mathbb{Z}[q]$. So it is a polynomial, in $\mathbb{Z}[q]$ or at least in $\mathbb{Q}[q]$. Our argument works for $c_{n,m} \in \mathbb{Q}[q]$ too (Remark 4). If $c_{n,m}$ were allowed to be an arbitrary rational function, (1) would hold for any polynomial in place of $\gcd(F_n, F_m)$ and would say nothing.

The exceptional set E . The phrase “a correction factor from a finite exceptional set E depending only on primes $p \equiv \pm 2 \pmod{5}$ ” is vague. A charitable reading lets the correction factor divide out a few cyclotomic factors $\Phi_p(q)$, or integer factors p , attached to finitely many exceptional primes $p \equiv \pm 2 \pmod{5}$, or excuses finitely many exceptional indices. Under that reading the counterexample $(6, 3)$ could be explained away. There the missing factor is $1 + q = \Phi_2(q)$,

and $2 \equiv 2 \pmod{5}$. The counterexample (11, 55) of Section 4 is not affected. Its indices involve only the primes 5 and 11. The missing factor is all of F_{11} , which has degree 25 and no cyclotomic factor. No correction built from finitely many primes or cyclotomic polynomials can turn F_{11} into $\gcd(F_{55}, F_{11}) = 1$.

2 Definitions and small values

The Gaussian binomial coefficient is $\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{i=1}^k \frac{1-q^{n-k+i}}{1-q^i}$ for $0 \leq k \leq n$, and 0 otherwise. It satisfies the q -Pascal rule $\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{k+1} \begin{bmatrix} n \\ k+1 \end{bmatrix}_q$. Only the terms with $0 \leq j \leq n-1-j$ contribute to F_n . We get

$$\begin{aligned} F_3 &= \begin{bmatrix} 2 \\ 0 \end{bmatrix}_q + \begin{bmatrix} 1 \\ 1 \end{bmatrix}_q q = 1 + q, \\ F_6 &= \begin{bmatrix} 5 \\ 0 \end{bmatrix}_q + \begin{bmatrix} 4 \\ 1 \end{bmatrix}_q q + \begin{bmatrix} 3 \\ 2 \end{bmatrix}_q q^4 = 1 + (q + q^2 + q^3 + q^4) + (q^4 + q^5 + q^6) \\ &= 1 + q + q^2 + q^3 + 2q^4 + q^5 + q^6. \end{aligned}$$

As a sanity check, $F_n(1) = \sum_j \binom{n-1-j}{j}$ is the Fibonacci number 1, 1, 2, 3, 5, 8, $\dots$. These are the classical q -Fibonacci polynomials of Schur, Carlitz and Andrews [1, 2].

Lemma 1. $F_3(-1) = 0$ and $F_6(-1) = 2$. Moreover $F_6 = F_3 \cdot (-1 + 2q - q^2 + 2q^3 + q^5) + 2$.

Proof. Direct computation. $F_6(-1) = 1 - 1 + 1 - 1 + 2 - 1 + 1 = 2$. □

3 The gcd of F_6 and F_3 in $\mathbb{Z}[q]$

Proposition 2. 1 is a greatest common divisor of F_6 and F_3 in $\mathbb{Z}[q]$. That is, every $D \in \mathbb{Z}[q]$ dividing both F_6 and F_3 divides 1, i.e. $D = \pm 1$.

Proof. Write $DA = F_3 = 1 + q$ and $DB = F_6$ with $A, B \in \mathbb{Z}[q]$. Degrees add in $\mathbb{Z}[q]$, because the top coefficient of a product is the product of the top coefficients and $\mathbb{Z}$ has no zero divisors. Hence $\deg D + \deg A = 1$, so D or A is constant.

If $A = a$ is constant, then $D(0)a = 1$ gives $a \neq 0$, and $D(-1)a = F_3(-1) = 0$ gives $D(-1) = 0$. But then $F_6(-1) = D(-1)B(-1) = 0$, contradicting $F_6(-1) = 2$ (Lemma 1).

If $D = d$ is constant, then $d \cdot A(0) = 1$ forces $d = \pm 1$, so $D \cdot d = 1$ and $D \mid 1$. □

Theorem 3. For $(n, m) = (6, 3)$ the clause (1) fails. Conjecture 00000007672 is false.

Proof. $\gcd(6, 3) = 3$ and $F_3 = 1 + q$. By Proposition 2 the gcd of F_6 and F_3 in $\mathbb{Z}[q]$ is ± 1 , which is unique up to units. If $\pm 1 = (1 + q)c$ for a polynomial c , then evaluating at $q = -1$ gives $\pm 1 = 0$, a contradiction. More generally, no divisor G of F_6 is a multiple of F_3 . If $G = F_3c$ and $GH = F_6$, then $2 = F_6(-1) = F_3(-1)c(-1)H(-1) = 0$. □

Remark 4. The same evaluation argument works if the correction factor c is allowed to lie in $\mathbb{Q}[q]$, or even to be any rational function without a pole at $q = -1$. Over $\mathbb{Q}[q]$ the gcd is also 1, because $F_6 \equiv 2 \pmod{1+q}$.

Remark 5 (The family $(3, 3k)$). The polynomials F_n satisfy Schur's recurrence $F_n = F_{n-1} + q^{n-2}F_{n-2}$ for $n \geq 3$ [2]. To see this, apply $\begin{bmatrix} N \\ j \end{bmatrix}_q = \begin{bmatrix} N-1 \\ j \end{bmatrix}_q + q^{N-j} \begin{bmatrix} N-1 \\ j-1 \end{bmatrix}_q$ with $N = n-1-j$. The first terms sum to F_{n-1} . In the second, put $i = j-1$: the exponent $j^2 + (n-1-2j)$ becomes $i^2 + n-2$, so these terms sum to $q^{n-2}F_{n-2}$. Put $f_n = F_n(-1)$, so $f_0 = 0$ and $f_1 = f_2 = 1$. Then $f_{2k} = f_{2k-1} + f_{2k-2}$ and $f_{2k+1} = f_{2k} - f_{2k-1} = f_{2k-2}$. Hence $f_{2k} = f_{2k-2} + f_{2k-4}$ for $k \geq 2$, with $f_2 = f_4 = 1$, so $f_{2k} = \text{Fib}_k$ and $f_{2k+1} = \text{Fib}_{k-1}$. In particular $f_n > 0$ for all $n \geq 1$ with $n \neq 3$. Thus $1+q = F_3$ divides no F_{3k} with $k \geq 2$, and the argument of Theorem 3 refutes (1) for every pair $(3, 3k)$, $k \geq 2$. (Lean checks f_{3k} for $k = 2, \dots, 6$.) All of these pairs share the missing factor $\Phi_2 = 1+q$, so this family alone does not rule out the cyclotomic reading of E . The next section does.

4 A counterexample avoiding the exceptional primes

Theorem 6. $\gcd(11, 55) = 11$, and no divisor of F_{55} in $\mathbb{Z}[q]$ is a multiple of F_{11} . In fact $\gcd(F_{55}, F_{11}) = 1$ in $\mathbb{Z}[q]$, while

$$F_{11} = 1 + q + q^2 + q^3 + 2q^4 + 2q^5 + 3q^6 + 3q^7 + 4q^8 + 5q^9 + 5q^{10} + 5q^{11} + 6q^{12} + \dots + q^{25}$$

is monic of degree 25 and is divisible by no cyclotomic polynomial.

Proof. From the definition, $F_{11}(-1) = 3$ and $F_{55}(-1) = 121393 = \text{Fib}_{26}$ (also from $f_{2k+1} = \text{Fib}_{k-1}$ above). If $G = F_{11}c$ and $GH = F_{55}$ with $c, H \in \mathbb{Z}[q]$, then $121393 = 3c(-1)H(-1)$. This is impossible because $121393 \equiv 1 \pmod{3}$. Since $c \in \mathbb{Q}[q]$ with $F_{11}c \in \mathbb{Z}[q]$ forces $c \in \mathbb{Z}[q]$ by Gauss's lemma (F_{11} is monic), the same holds over $\mathbb{Q}[q]$.

The remaining statements are computational and are checked in `verify.py`. Both F_{11} and F_{55} are monic, so a nontrivial common factor over $\mathbb{Q}$ would be a monic integer polynomial, and its reduction modulo the prime $2^{61} - 1$ would divide both reductions. The Euclidean algorithm modulo $2^{61} - 1$ gives a gcd of degree 0, so $\gcd(F_{55}, F_{11}) = 1$. A cyclotomic factor Φ_k of F_{11} would need $\varphi(k) \leq 25$, hence $k \leq 1250$ (since $\varphi(k) \geq \sqrt{k}/2$). None of the 53 such Φ_k divides F_{11} . $\square$

Thus for $(n, m) = (11, 55)$ the gcd is $1 = F_{11} \cdot c$ only for $c = 1/F_{11}$, which is not a polynomial. It is also not a correction made of finitely many primes $\equiv \pm 2 \pmod{5}$ or of their cyclotomic factors. Failures are common: $F_d \nmid F_{2d}$ for every $3 \leq d \leq 30$ except $d = 5$.

Remark 7 (Infinitely many indices). Even a reading in which E is an arbitrary finite set of exceptional indices does not rescue the conjecture. Let $d = 2a + 1$ be odd. By the formulas above, $F_d(-1) = \text{Fib}_{a-1}$ and $F_{5d}(-1) = F_{2(5a+2)+1}(-1) = \text{Fib}_{5a+1}$. For $a-1 \geq 3$ we have $\text{Fib}_{a-1} \mid \text{Fib}_{5a+1}$ if and only if $(a-1) \mid (5a+1)$, which holds if and only if $(a-1) \mid 6$. So for every odd $d \geq 11$ with $d \neq 15$, $F_d(-1) \nmid F_{5d}(-1)$. Hence no divisor of F_{5d} is a multiple of $F_d = F_{\gcd(d, 5d)}$, and (1) fails for $(d, 5d)$. This gives infinitely many failures with unbounded gcd, including every prime $d \equiv \pm 1 \pmod{5}$. (`verify.py` checks this for odd $d < 200$.)

5 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project that uses only the core library (no `Mathlib`).

- Polynomials in $\mathbb{Z}[q]$ are coefficient lists. `PEq` means equal coefficients, `Dvd a b` is $\exists c, a \cdot c = b$ in $\mathbb{Z}[q]$, and `IsGcd G a b` is the standard definition of a greatest common divisor (a common divisor that every common divisor divides).

- `gauss` defines Gaussian binomials by the q -Pascal rule. `gauss_product_formula` checks $\begin{bmatrix} n \\ k \end{bmatrix}_q (q; q)_k (q; q)_{n-k} = (q; q)_n$ for all $k \leq n \leq 7$. `qfib n` is $\sum_{j < n} \begin{bmatrix} n-1-j \\ j \end{bmatrix}_q q^{j^2}$ exactly as in the conjecture. `qfib_at_one` checks $F_n(1) = \text{Fib}_n$ for $n \leq 15$.
- `common_divisor_dvd_one` and `isGcd_one` formalize Proposition 2. They use the degree argument (`coeff_mull_top`, `const_or_const`) and evaluation (`evalL_mull`, `evalL_congr`).
- `GcdClause` is the clause (1): for all $n, m \geq 1$ and every gcd G of F_n, F_m in $\mathbb{Z}[q]$, $G = F_{\gcd(n,m)} \cdot c$ for some $c \in \mathbb{Z}[q]$. `conjecture_00000007672_false` proves $\neg \text{GcdClause}$ using (6, 3). `no_divisor_of_F6_is_multiple_of_F3` proves the stronger statement from the proof of Theorem 3.
- Since a gcd in $\mathbb{Z}[q]$ always exists and is unique up to sign, the universal form (“every gcd is $F_{\gcd(n,m)} \cdot c$ ”) and the existential form (“some gcd is $F_{\gcd(n,m)} \cdot c$ ”) of (1) are equivalent. The Lean project refutes both. `GcdClauseExists` is the existential form: for all $n, m \geq 1$ there are a gcd G and $c \in \mathbb{Z}[q]$ with $G = F_{\gcd(n,m)} \cdot c$. `conjecture_00000007672_false_exists_6_3` and `conjecture_00000007672_false_11_55` prove $\neg \text{GcdClauseExists}$ using (6, 3) and (11, 55); the latter goes via `no_divisor_of_F55_is_multiple_of_F11`. $F_{55}(-1)$ is computed with a row-by-row evaluation of Gaussian binomials (`row`, `fastF`), proved equal to evaluating the polynomial (`evalL_qfib`).
- Lean takes the correction factor in $\mathbb{Z}[q]$. The extension to $\mathbb{Q}[q]$ (Gauss’s lemma) and the facts that $\gcd(F_{55}, F_{11}) = 1$ and that F_{11} has no cyclotomic factor are shown in this report and in `verify.py`, not in Lean.

`lake build` succeeds. There is no `sorry`, no `native_decide`, and no added axioms. `#print axioms` shows at most `propext`, `Classical.choice` and `Quot.sound`. The numerical facts are proved by `decide`. `qfib_mult3_at_neg_one` and `qfib_55_at_neg_one` use `decide +kernel`, which is checked by the Lean kernel.

6 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py          # standard library only
pdflatex report.tex
```

`verify.py` computes Gaussian binomials from the product formula by exact division, which is independent of the q -Pascal rule used in Lean. It recomputes F_n , finds $\gcd(F_6, F_3) = 1$ by the Euclidean algorithm in $\mathbb{Q}[q]$ (and in $\mathbb{Z}[q]$ by Gauss’s lemma, since F_3 is primitive), and lists all 76 pairs $n < m \leq 30$ for which $F_{\gcd(n,m)} \nmid F_n$ or $F_{\gcd(n,m)} \nmid F_m$. It also performs all the checks for (11, 55) in Theorem 6 and verifies Schur’s recurrence for $n \leq 30$.

References

- [1] L. Carlitz, Fibonacci notes 3: q -Fibonacci numbers, *Fibonacci Quart.* 12 (1974), 317–322.
- [2] G. E. Andrews, Fibonacci numbers and the Rogers–Ramanujan identities, *Fibonacci Quart.* 42 (2004), 3–19.